Namrata Deka

PhD Student · Machine Learning

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Summary

Ph.D. student in Machine Learning at Carnegie Mellon University researching multimodal representation learning, controllable generation, and decision-making. My research develops structured representations that connect language, sensory signals, states and actions for robust generation and planning. Seeking research internships in multimodal learning, generative modeling and embodied/decision-making systems. Anticipated graduation: 08/2028.

Education

Carnegie Mellon University *Pittsburgh, U.S.A.*
Ph.D., Machine Learning *2023 - Present*

- Advisors: Dr. Jeff Schneider, Dr. Kun Zhang

University of British Columbia *Vancouver, Canada*
M.Sc., Computer Science *2020 - 2023*

- Advisor: Dr. Danica J. Sutherland
- Thesis: Kernel Methods for Invariant Representation Learning: Enforcing Fairness and Conditional Independence

Indraprastha Institute of Information Technology *New Delhi, India*
B.Tech., Computer Science and Engineering *2013 - 2017*

- Honors thesis/undergrad research advisors: Dr. Saket Anand, Dr. Sanjit K. Kaul

Experience

Apple *Cupertino, U.S.A.*
Machine Learning Intern *May 2026 - Aug. 2026*

- Developed conditional and unconditional generative models for wearable biosignals using diffusion models, flow matching, and foundation-model representations.
- Designed a comprehensive evaluation framework for synthetic biosignals, assessing realism, diversity, controllability, and downstream utility for data augmentation.

Carnegie Mellon University – Dept. of Machine Learning *Pittsburgh, U.S.A.*
Ph.D. Research Assistant *2023 - Present*

- Develop structured multimodal representations linking states, actions, sensory observations, and language for controllable generation and decision-making.
- Study latent concept alignment and abstraction for learning transferable dynamics and planning representations across modalities.
- Develop multimodal scientific AI systems that integrate experimental time series, control trajectories, and text logs to support experiment understanding and planning.

Borealis AI *Vancouver, Canada*
ML Research Intern *Sep. 2022 - Jan. 2023*

- Developed slice-discovery and model-diagnostic methods to identify systematic performance failures of ML models across underrepresented subpopulations; built evaluation tooling to characterize and guide mitigation of discovered failure modes.

University of British Columbia – Dept. of Computer Science *Vancouver, Canada*
M.Sc. Research Assistant *2020 - 2023*

- Developed a conditionally invariant representation learning model using kernel-based conditional independence measures; published at ICLR 2023 (Oral/Top 5%).
- Developed a fair representation learning model with statistical testing guarantees; published at AISTATS 2023.

École Polytechnique Fédérale de Lausanne (EPFL) *Lausanne, Switzerland*

Computer Vision Research Intern, Summer@EPFL *Jun. 2021 - Aug. 2021*

- Researched and developed methods for articulated object modeling from video using 3D optical flow and depth estimation, including end-to-end training, inference, and systematic ablation studies.

Wadhwani Institute for AI *Mumbai, India*

AI Research Fellow *Aug. 2018 - Aug. 2020*

- Built a smartphone-based 3D vision system for low-birth-weight screening in resource-constrained public-health settings.
- Developed a novel method to reconstruct 3D infant meshes at metric scale.

Microsoft Research *Bengaluru, India*

Applied ML Intern *Jan. 2018 - Jul. 2018*

- Developed a GDPR-oriented PII detection system targeting rare sensitive categories, improving robustness through targeted data curation and systematic error analysis.

Publications

Peer-Reviewed Conferences

P. Zhu, S. Xie, Z. Li, Y. Shen, **N. Deka**, H. Shrivastava, G. Chen, K. Zhang. *MoVA: Learning Asymmetric Dual Projections for Modular Long Video-Text Alignment*, in the 19th European Conference on Computer Vision (**ECCV**), 2026.

Y. Shen, P. Zhu, Z. Li, S. Xie, **N. Deka**, Z. Liu, Z. Tang, G. Chen, K. Zhang. *Controllable Video Generation with Provable Disentanglement*, in the 14th International Conference on Learning Representations (**ICLR**), 2026.

Z. He, R. Pogodin, Y. Li, **N. Deka**, A. Gretton, D. Sutherland. *On the Hardness of Conditional Independence Testing in Practice*, in the 39th Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2025 (Spotlight).

R. Pogodin*, **N. Deka***, Y. Li*, D.J. Sutherland, V. Veitch, A. Gretton. *Efficient Conditionally Invariant Representation Learning*, in the 11th International Conference on Learning Representations (**ICLR**), 2023 (Oral/Top 5%). *Equal Contribution.

N. Deka, D.J. Sutherland. *MMD-B-Fair: Learning Fair Representations with Statistical Testing*, in the 26th International Conference on Artificial Intelligence and Statistics (**AISTATS**), 2023.

Submitted / Under Review

K. Zhang, J. Sun, Y. Li, I. Ng, **N. Deka**, S. Xie. *Structural Extrapolative Data Generation: Hardness, Feasibility and Formal Conditions*, 2026.

Workshops & Technical Reports

Y. Khandelwal, M. Arvind, S. Kumar, A. Gupta, S.K. Danisetty, P. Bagad, A. Madan, M. Lunayach, A. Annavajjala, A. Maiti, S. Jain, A. Dalmia, **N. Deka**, J. White, J. Doshi, A. Kanazawa, R. Panicker, A. Raval, S. Rana, M. Tapaswi. *NurtureNet: A Multi-task Video-based Approach for Newborn Anthropometry*, in the 7th **CVPR** Workshop on Computer Vision for Physiological Measurements (CVPM), 2024 (Best Paper).

N. Deka, D. Sutherland. *Learning Privacy-Preserving Deep Kernels with Known Demographics*, in the 36th **AAAI** Conference on Artificial Intelligence. Workshop on Privacy-Preserving Artificial Intelligence (PPAI), 2022.

D. Sutherland, **N. Deka**. *Unbiased estimators for the variance of MMD estimators*. Technical report 2022 (arXiv:1906.02104).

Technical Skills

Programming / Systems: Python; Java; C; C++; Slurm; Docker

ML: PyTorch; TensorFlow; ROS; Flow Matching; Diffusion Models; LLMs; Causality; Representation Learning; Reinforcement Learning; Kernel Methods; Hypothesis Testing

Service & Teaching

Peer Review: NeurIPS 2024 (CRL Workshop), 2026; AISTATS 2023, 2026 (CauScale Workshop); CLeaR 2026; ECCV 2026

Teaching Assistant: Machine Learning in Practice (CMU); Intelligent Systems (UBC); Computer Vision and Advanced Programming (IIIT-D)